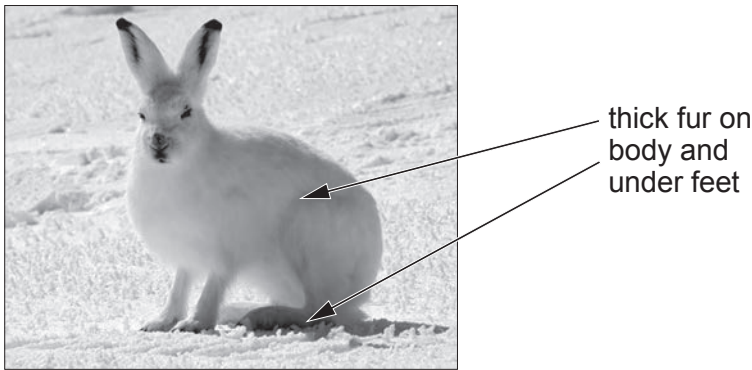


Answer all questions

1. The photograph shows an arctic hare (*Lepus arcticus*).



(a) The table below shows some classes of vertebrate animals.

Class	Features of animals	Examples
fish	skin with scales	goldfish, cod
reptiles	skin with scales	crocodiles, snakes
birds	skin with feathers	eagle, pigeon
mammals	skin with hair	horse, cat

(i) Use the information above and your own knowledge to complete the classification of the arctic hare. [3]

Kingdom animal

Phylum vertebrate

Class

Genus

Species

(ii) Why do scientists use scientific names for living organisms? [1]

.....



(b) Read the information about the arctic hare and use it to answer the questions which follow.

The arctic hare lives in northern countries where the climate is very cold. The average life expectancy is five years. Each has a body mass of 5 – 7 kg and 20% of this is a layer of fat under the skin, which helps to reduce heat loss. They live in large groups of about 200 individuals and huddle together while sleeping, to retain heat.

Arctic hares feed on berries, twigs and moss which they can dig out of snow with their strong feet. Only a few other species eat the same food as the hares but many others, such as foxes, wolves and lynx kill and eat large numbers of hares.

(i) Complete the table below by writing ‘true’ or ‘false’ for each of the statements about the arctic hare. [3]

Statement	True or False
All arctic hares live for at least five years.	
Arctic hares have little inter-specific competition for food.	
The population size of arctic hares is affected by predation.	
The layer of fat under the skin raises the body temperature.	

(ii) Explain why arctic hares are at a high risk of spreading disease while they sleep. [2]

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(c) Give **two** reasons why the arctic hare’s coat increases its chances of survival. [2]

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